

LE XUAN HUNG, PhD

Structural Bridge Engineer ▪ Engineering Software Developer

☎ +82 10 9941 0201 • ✉ lxhungdp@gmail.com • in [lxhungdp](#) • Seoul, South Korea

🐙 github.com/lxhungdp • 🌐 lehung.dev

U.S. status: Green Card expected in 2026; no employer visa sponsorship required.

PE (U.S.): Licensure expected in 2026; PE exam scheduled and preparation underway.

Professional Summary

- Bridge engineer with 10+ years of experience designing and analysing steel, concrete, truss, and cable-supported bridges. Also worked on other steel structures and temporary works for bridge construction.
- Strong foundation in structural mechanics and finite element methods. Experienced in global and detailed modelling, nonlinear and dynamic analysis, staged-construction analysis, and seismic response of bridges.
- Developing bridge-design software since 2018. Use C/C++ for structural-analysis solvers, C#/.NET for desktop applications, and Python and TypeScript for engineering automation and web tools.

Core Skills

- **Bridge design:** Steel bridges: steel I-girder and steel-concrete composite box-girder bridges. Concrete bridges: cast-in-place and precast girder bridges; segmental bridges built by balanced-cantilever and span-by-span methods. Cable-supported bridges: cable-stayed and suspension bridges. Arch bridges and truss bridges.
- **Other structures:** Steel structures, steel connections, and anchorages; temporary works for bridge construction, including scaffolding, falsework, formwork, and form travellers.
- **Development:** C and C++ for numerical solvers; C#/.NET and Visual Basic/VB.NET for desktop applications; Python; JavaScript/TypeScript with React and Node.js; VBA and MATLAB for engineering calculations; AutoLISP and AutoCAD .NET API for CAD automation; SQL and MongoDB; Git for version control.

Selected Engineering Software

Project demos and details: 🐙 github.com/lxhungdp 🌐 lehung.dev.

- **PUS Pro Program**

Structural analysis and design software for steel-concrete composite box-girder bridges, with an in-house FEM solver and design checks.

- **Truss Designer**

Analysis and design of steel and timber trusses.

- **Reinforced Concrete Column Designer**

Biaxial compression-bending design and capacity checks for reinforced-concrete columns.

- **Steel Bridge Designer**

Web tool under development for steel I-girder and box-girder bridge design.

- **Anchor Design**

Tool for the design and verification of structural anchors.

- **Bridge Alignment**

Horizontal and vertical alignment design for bridges, supporting geometry calculations and site coordination.

- **CastControl**

Web application for segment casting and match-casting geometry. Connects field survey data to design review and automated tolerance checks.

Design checks in these tools are based on AASHTO LRFD, Eurocodes, and KDS where applicable.

Professional Experience

ENVICO Co., Ltd.

Deputy Project Manager & Engineering Software Developer

Seoul, Korea

Sep 2024 – Present

Key responsibilities

- Lead engineering software development to convert the company's bridge-design workflows into software. Use AI-assisted development and check the engineering calculations and outputs.
- Perform complex structural analysis and design calculations for critical bridge structures, including seismic, nonlinear, and staged-construction analysis.

Selected projects

- *South Commuter Railway (Philippines) – Prestressed segmental bridges.* Developed CastControl for segment geometry and tolerance checks, reducing manual engineering effort by up to 95%.
- *Saemangeum Ring Road TK project (Korea) – Composite steel I-girder spans and a cable-stayed bridge with a circular pylon.* Performed global modelling, detailed structural analysis, seismic evaluation, design calculations, and documentation for review.

Wooryung Construction Co., Ltd.

Senior Structural Engineer & .NET Developer

Seoul, Korea

Sep 2018 – Sep 2024

Key responsibilities

- Developed .NET-based applications to automate steel box-girder bridge design and verification, including calculation sheets, technical reports, and quantity take-offs.
- Analysed steel bridges and construction temporary works, including trusses, scaffolding, formwork, erection supports, and steel connections.

Selected projects

- *PUS Pro program – Prestressed composite U-type steel box-girder design and verification.* Developed a .NET desktop application for structural analysis and design checks, with calculation sheets, quantities, and technical reports.
- *Hyeongok 2nd Bridge, Ihwa–Samgye Road widening (Section 2) – Steel composite box girder.* Performed structural modelling and analysis, steel connection and formwork design, calculations, drawings, and review documentation.

Hung Nghiep Construction Consulting Co., Ltd.

Structural Engineer

HCMC, Vietnam

Aug 2007 – Aug 2010

Key responsibilities

- Conducted structural and staged-construction analysis for balanced-cantilever, precast and cast-in-place concrete, and temporary steel truss bridges.
- Used VBA, AutoLISP, and Mathcad to automate bridge-design calculations and prepare calculation sheets.

Selected project

- *An Nghia Bridge – Can Gio, HCMC.* Prepared detailed design and construction-stage analysis for balanced-cantilever construction.

Education

2012–2018 PhD Civil Engineering, Kyung Hee University

Research on bridge deflection and vibration reliability; peer-reviewed publications.

2010–2012 MEng, Kyung Hee University

2002–2007 BSc Civil Engineering, University of Communications and Transport, Vietnam